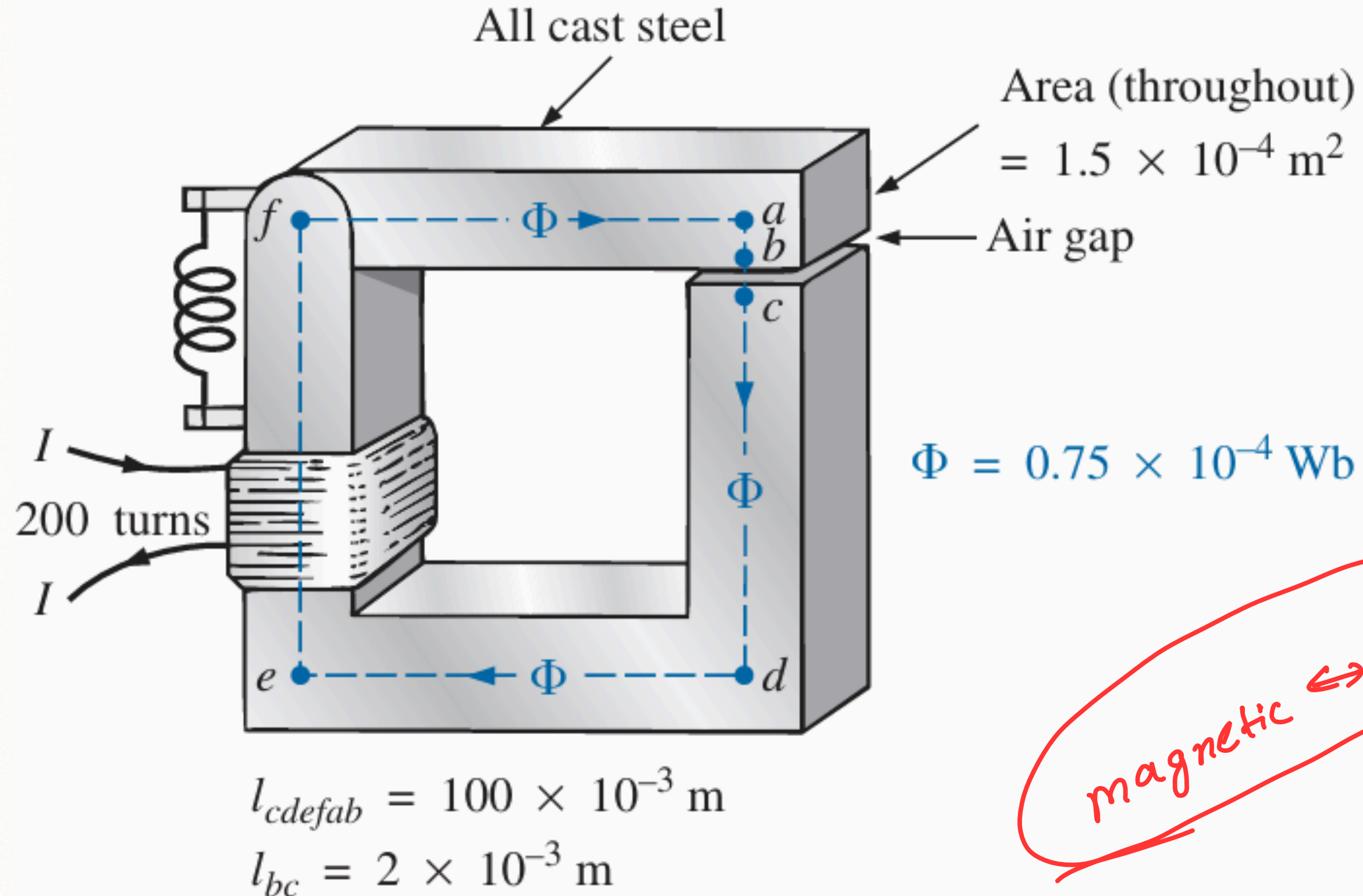


Muslim Engineers Academy

MAGNETIC CIRCUIT

MAGNETIC CIRCUIT



magnetic $\leftrightarrow$ Electric

MAGNETIC FIELD FLUX DENSITY

$$B = \frac{\Phi}{A}$$

$$B = \text{Wb/m}^2 = \text{teslas (T)}$$

$$\Phi = \text{webers (Wb)}$$

$$A = \text{m}^2$$

$$B = \frac{\Phi \rightarrow \text{wb}}{A \rightarrow \text{m}^2}$$

$$\text{wb/m}^2 \rightarrow \text{tesla}$$

$$B = \frac{\Phi}{A}$$

RELATIVE PERMEABILITY

The ratio of the permeability of the material to that of air is called the relative permeability

$$\mu_r = \frac{\mu}{\mu_o}$$

$$\mu_o = 4\pi \times 10^{-7} \text{Wb/A} \cdot \text{m}$$

$$\mu_r = \frac{\mu}{\mu_o}$$
$$\mu_o = 4\pi \times 10^{-7} \text{Wb/A} \cdot \text{m}$$

RELUCTANCE

R

The resistance of a material to the flow of charge (current) is determined for electric circuits by the equation

$$R = \rho \frac{l}{A} \quad (\text{ohms, } \Omega)$$

$$R = \frac{\rho l}{A}$$

$$\mathcal{R} = \frac{l}{\mu A}$$

The **reluctance** of a material to the setting up of magnetic flux lines in the material is determined by the following equation:

$$\mathcal{R} = \frac{l}{\mu A}$$

(rels, or At/Wb)

$$\mathcal{R} = \frac{l}{\mu A} \quad (12.4)$$

MAGNETOMOTIVE FORCE

$$\mathcal{F}_p = \text{ampere-turns (At)}$$

$$N = \text{turns (t)}$$

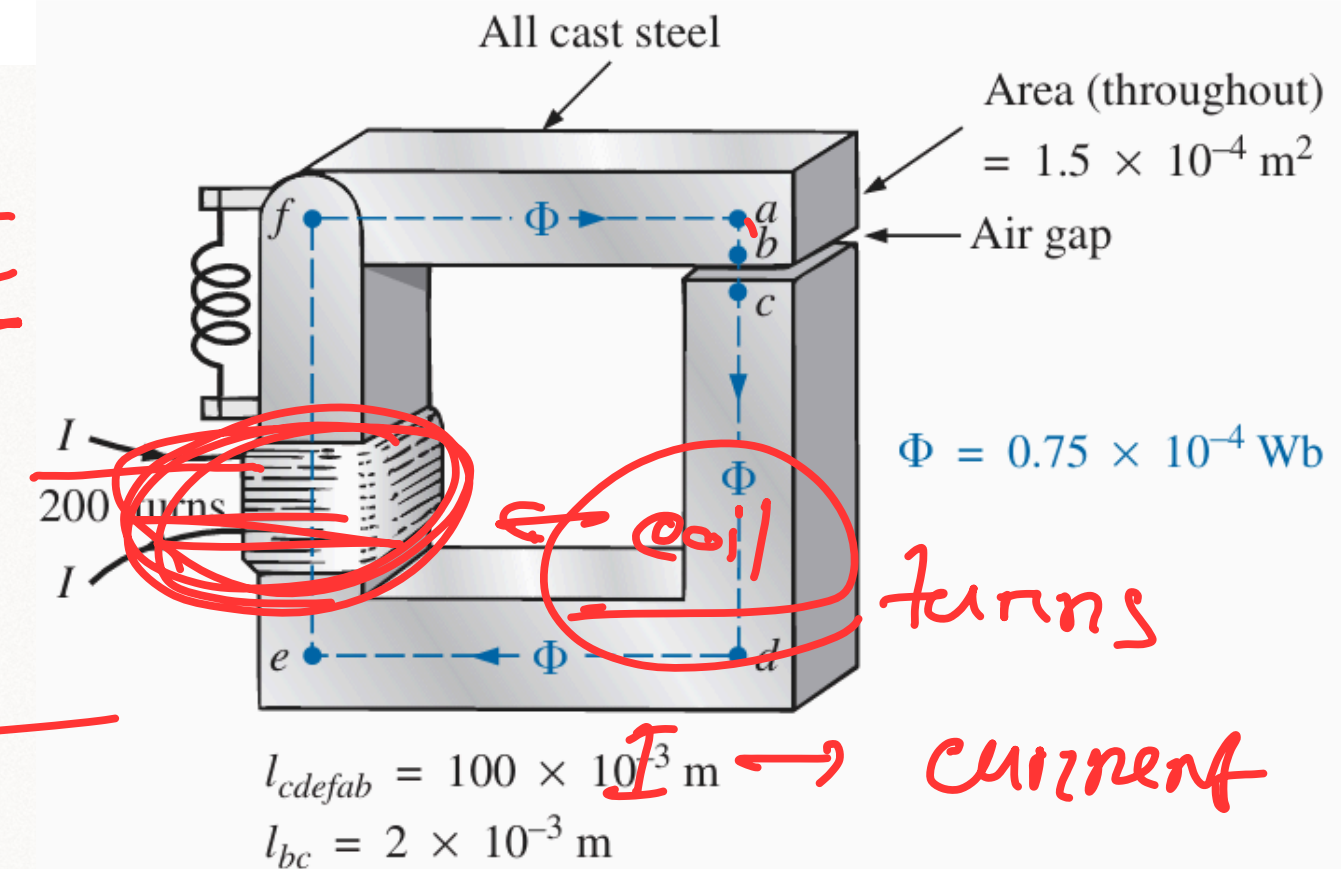
$$I = \text{amperes (A)}$$

$$\mathcal{F}_\phi = NI$$

$$f_0 = \frac{NI}{1 \uparrow}$$

$$J = N \frac{1}{2}$$

$$F = NI$$



MAGNETIZING FORCE

$$H = \frac{F}{l}$$

$$H = \frac{\mathcal{F}}{l}$$

$$H = \frac{NI}{l}$$

$$H = \frac{NI}{l}$$

$$H = \frac{NI}{l}$$

$$B = \mu H$$

magnetizing

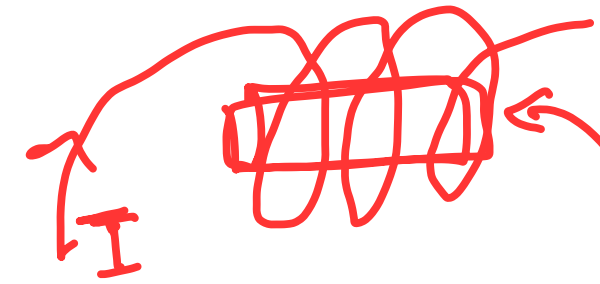
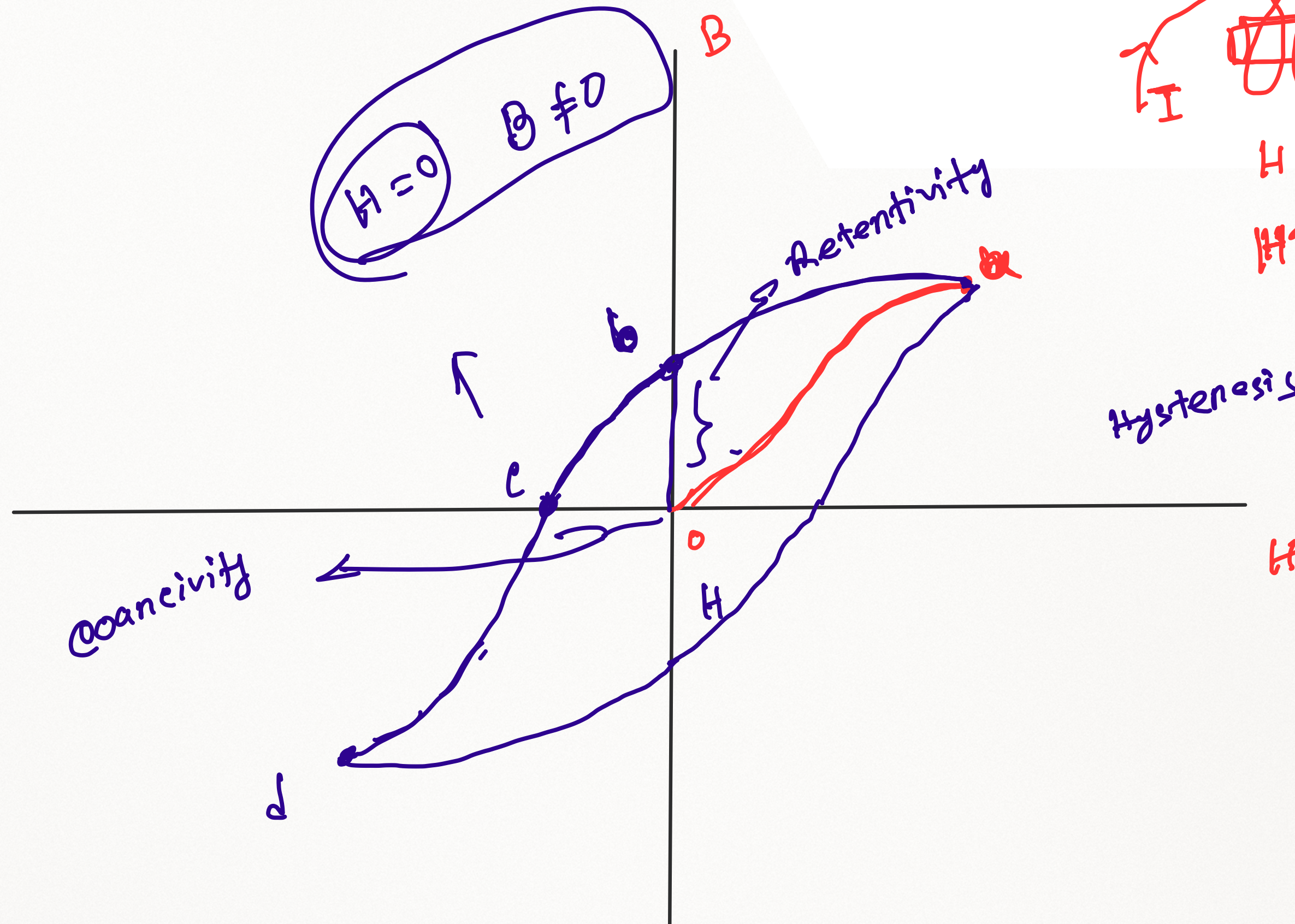
$$B = \mu H$$

$$F$$

$$F_{\text{ext}}$$

HYSTERESIS

→ Ferromagnetic



magnetism

$H=0$ $B=0$
 $H \uparrow \Rightarrow I \uparrow \Rightarrow B \uparrow$
 $I \uparrow$

$\textcircled{H}$ m
 $\uparrow$
 $B \propto \textcircled{H}$
 $\uparrow$

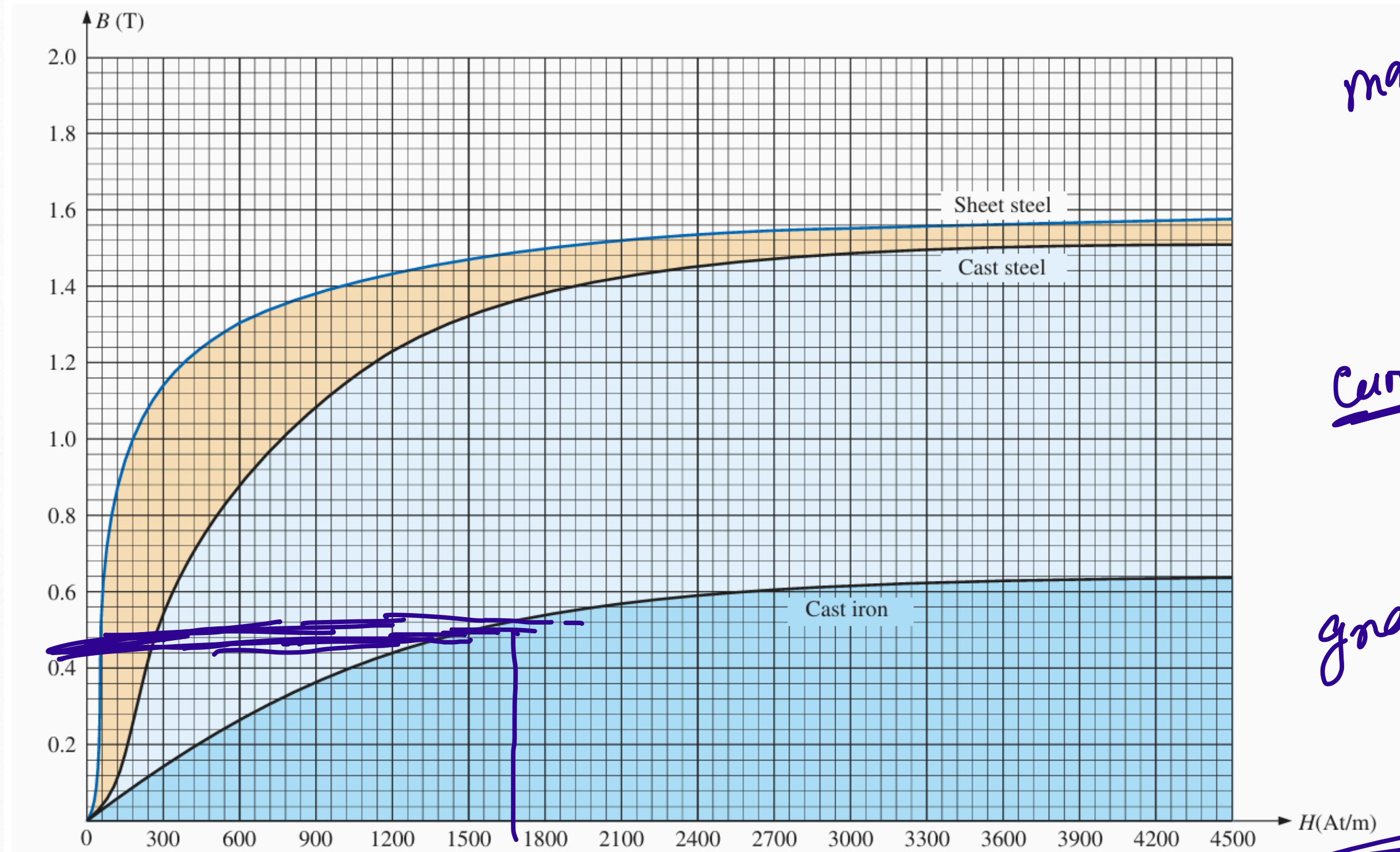
$B \neq 0$ $H = \frac{NI}{l}$

$\textcircled{\Phi}$ magnetism

HYSTERESIS



HYSTERESIS



materials cast iron

$$B = 0.4$$

$$H = ?$$

Curve

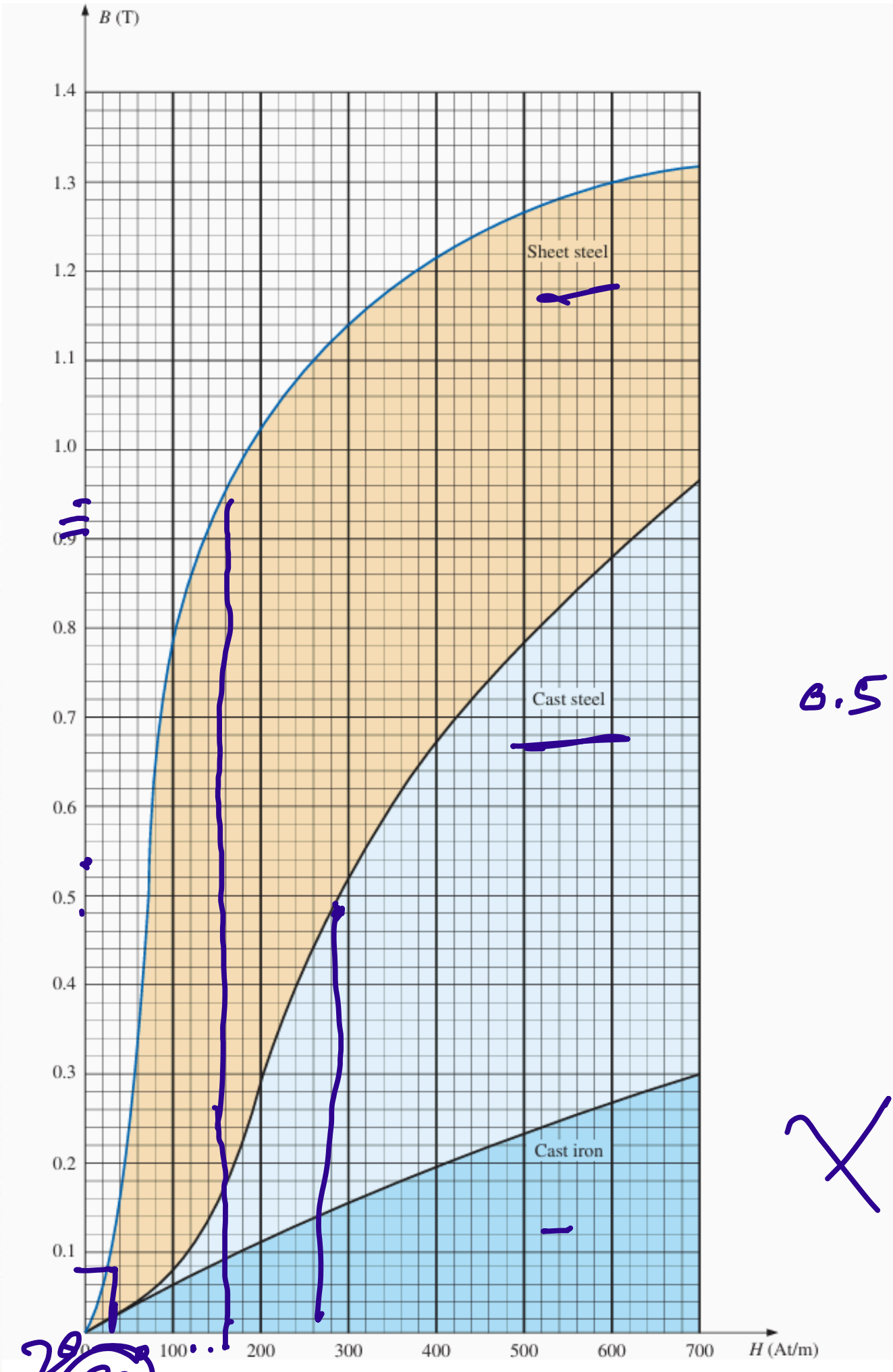
Cast steel

graph

$$\underline{H = 300}$$

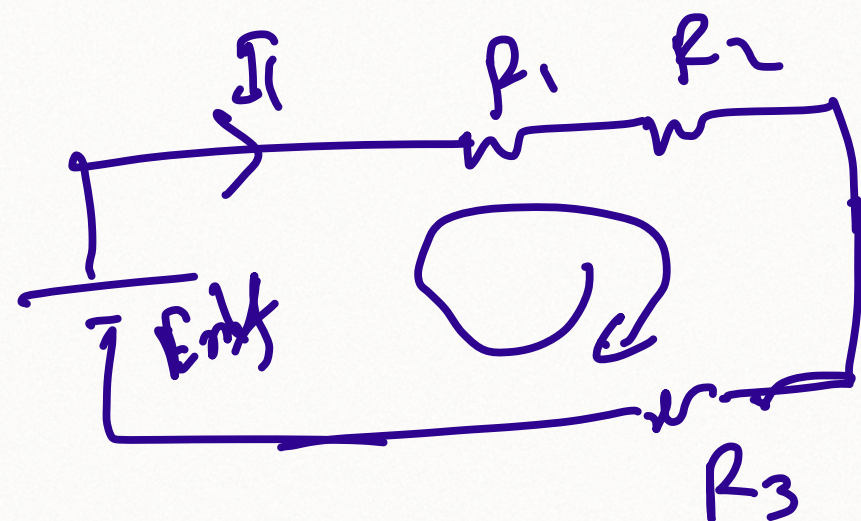
1600

HYSTERESIS



AMPÈRE'S CIRCUITAL LAW

$$\oint \mathbf{J} \cdot d\mathbf{l} = 0$$



$$-V + IR_1 + IR_2 + IR_3 = 0$$

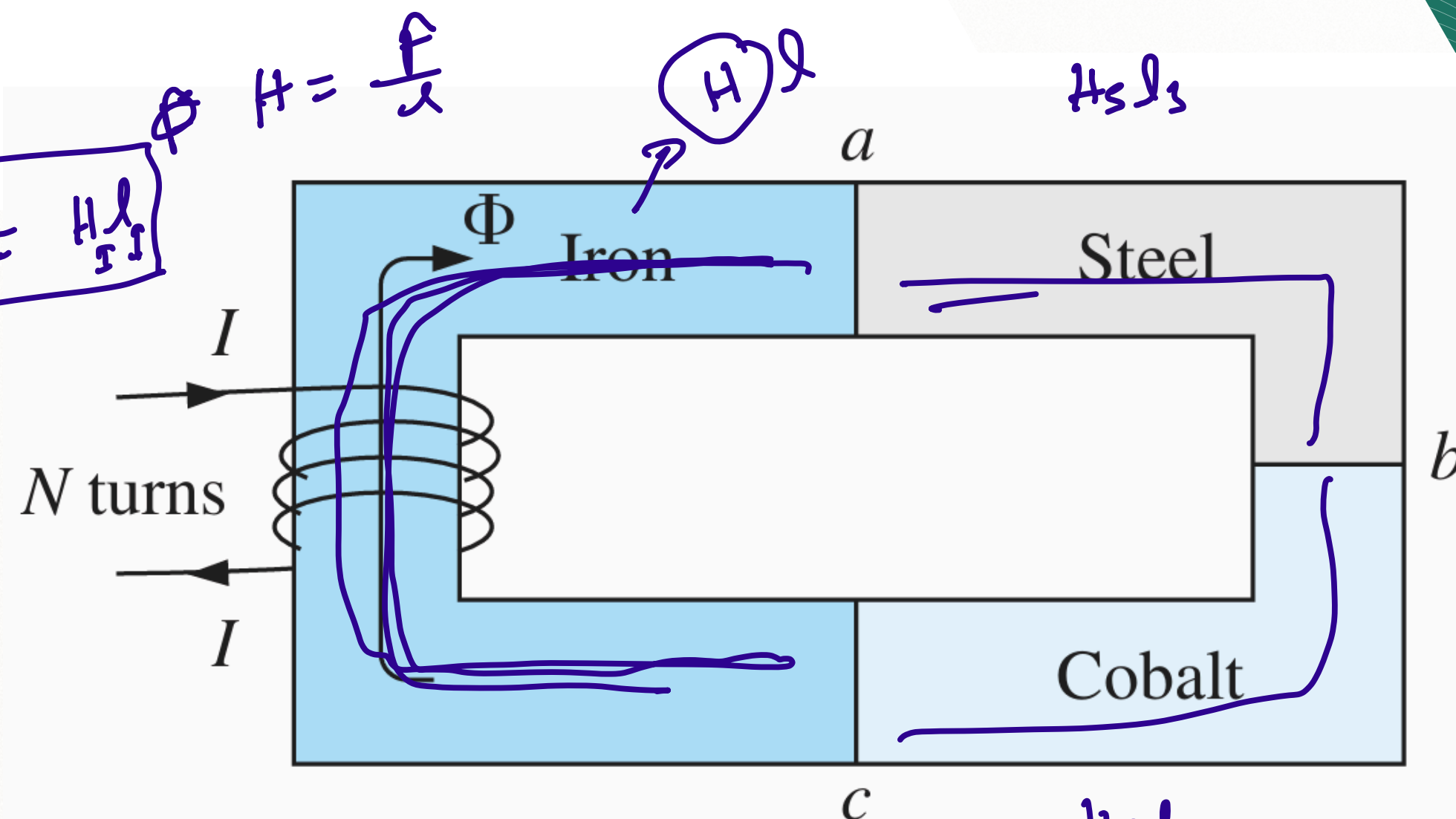
$$V - V_1 - V_2 - V_3 = 0$$

$$\sum V = 0$$

$$V - IR_1 + IR_2 + IR_3 = 0$$

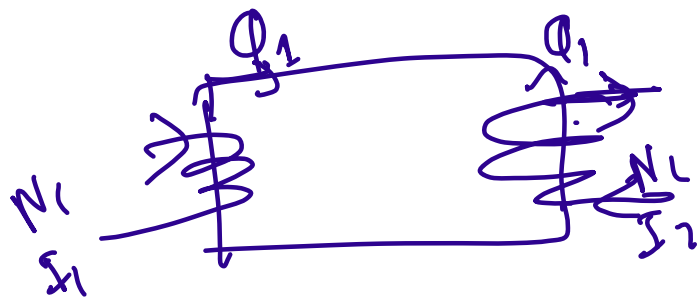
$$\sum V = 0$$

$$\oint \mathbf{H} \cdot d\mathbf{l} = \oint \mathbf{J} \cdot d\mathbf{l}$$

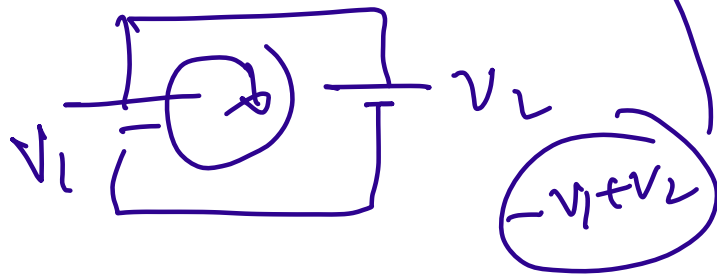


$$NI = H_1 l_1 + H_2 l_2 + H_3 l_3 + H_4 l_4$$

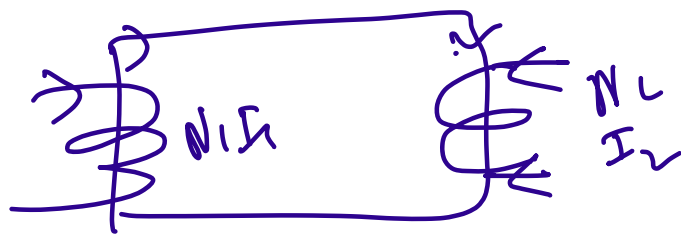
$$H_1 I - H_2 l_1 - H_3 I - H_4 l_2$$



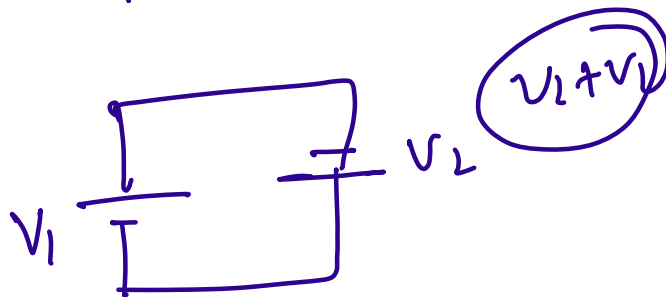
$$N_1 I_1 - N_2 I_2$$



$$-V_1 + V_2$$

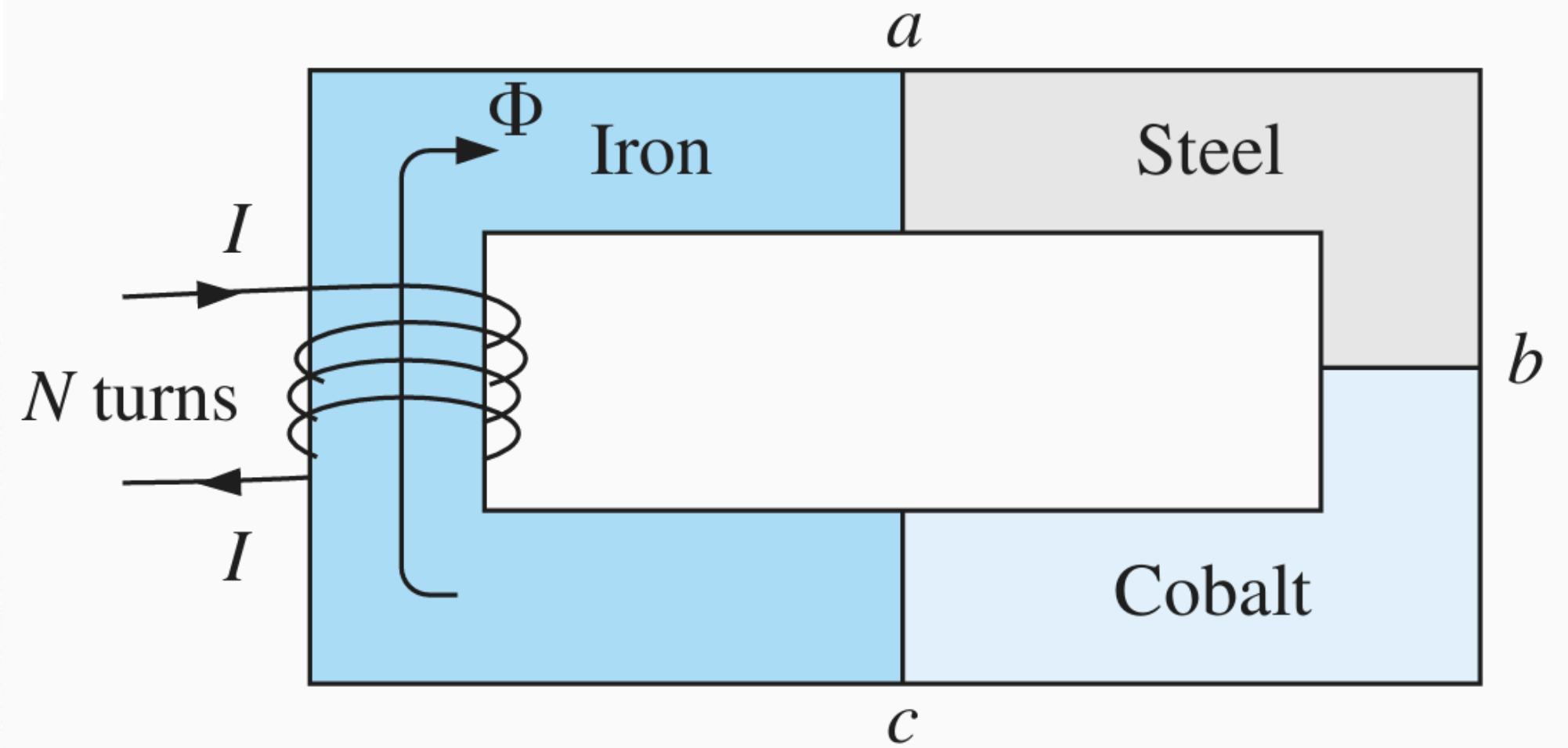


$$N_1 I_1 + N_2 I_2$$



$$V_1 + V_2$$

AMPÈRE'S CIRCUITAL LAW



12.8 FLUX Φ

If we continue to apply the relationships described in the previous section to Kirchhoff's current law, we find that the sum of the fluxes entering a junction is equal to the sum of the fluxes leaving a junction; that is, for the circuit in Fig. 12.11,

$$\Phi_a = \Phi_b + \Phi_c \quad (\text{at junction } a)$$

or

$$\Phi_b + \Phi_c = \Phi_a \quad (\text{at junction } b)$$

both of which are equivalent.

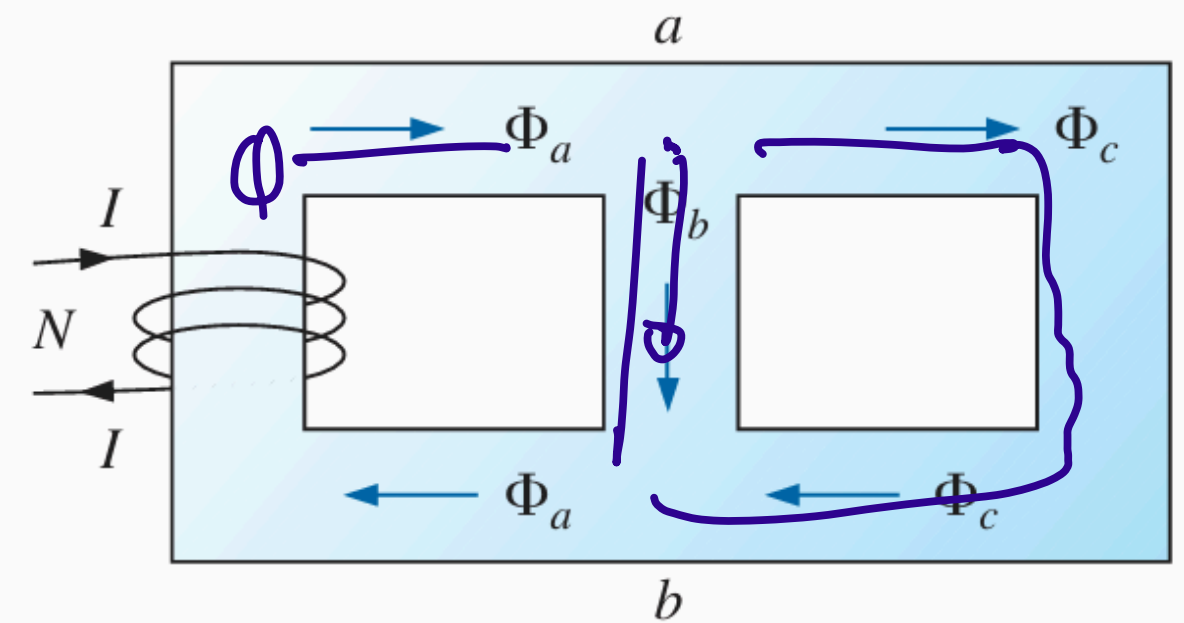


FIG. 12.11

Handwritten notes:

$$\mathcal{F} = NI$$

$$H = \frac{NI}{l} \quad Hl = NI$$

$$\beta = \frac{\Phi}{A}$$

$$\beta = \mu H$$

$$NI = H_1 l_1 + H_2 l_2$$

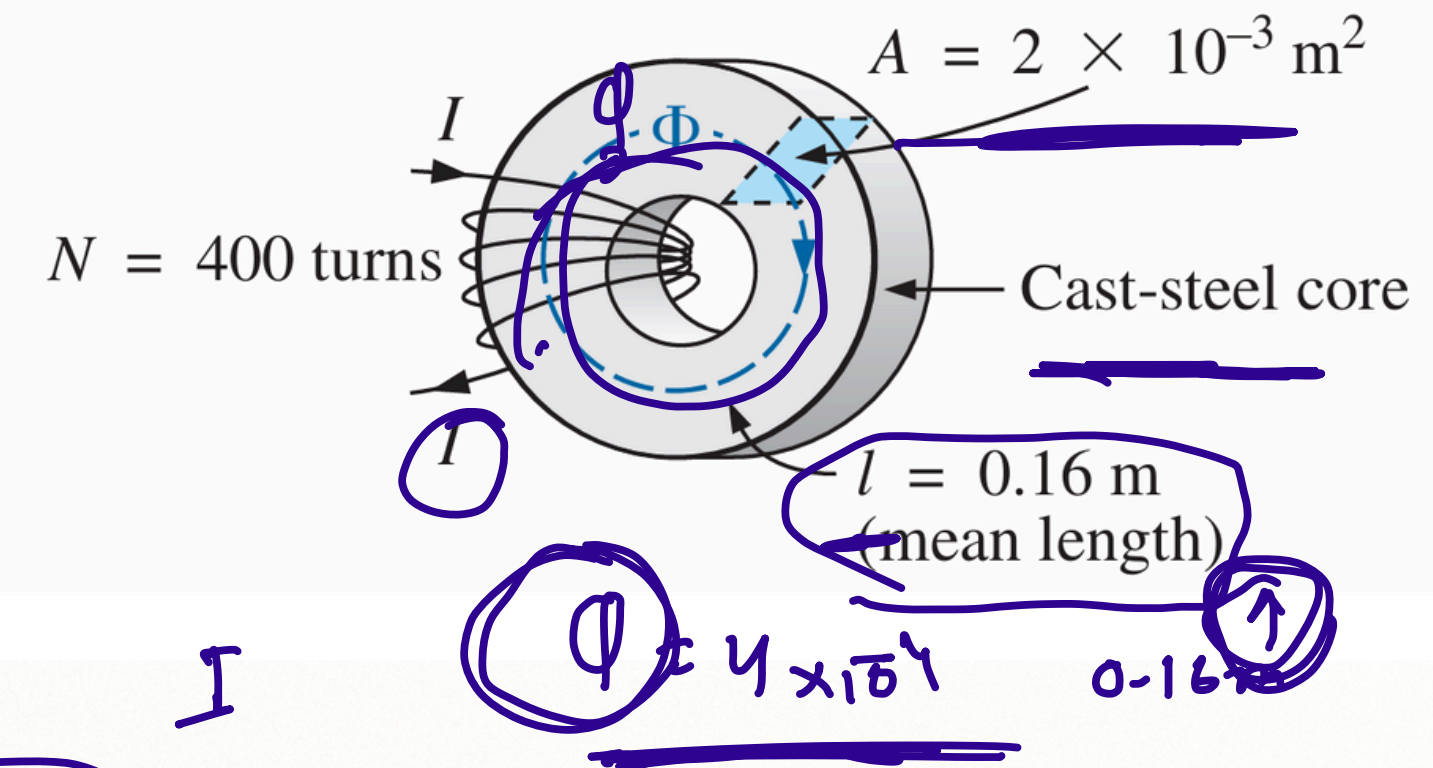
$$NI = H_1 l_1 + H_2 l_2 + H_3 l_3$$

$$\sum NI = \sum Hl$$

Short

EXAMPLE 12.1 For the series magnetic circuit in Fig. 12.12:

- Find the value of I required to develop a magnetic flux of $\Phi = 4 \times 10^{-4}$ Wb.
- Determine μ and μ_r for the material under these conditions.



$$B = \frac{\Phi}{A}$$
$$= \frac{4 \times 10^{-4}}{2 \times 10^{-3}} = 0.2 \text{ T}$$

$$H = 170 \text{ A/m}$$

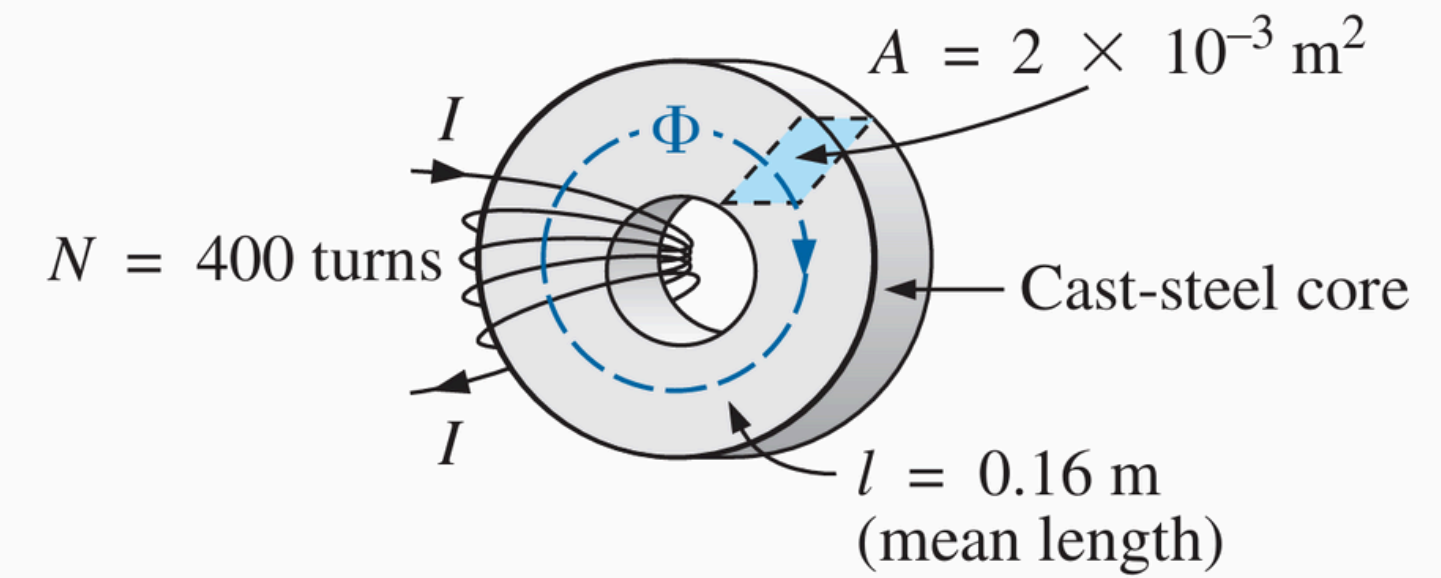
$$NI = Hl$$

$$I = \frac{Hl}{N}$$

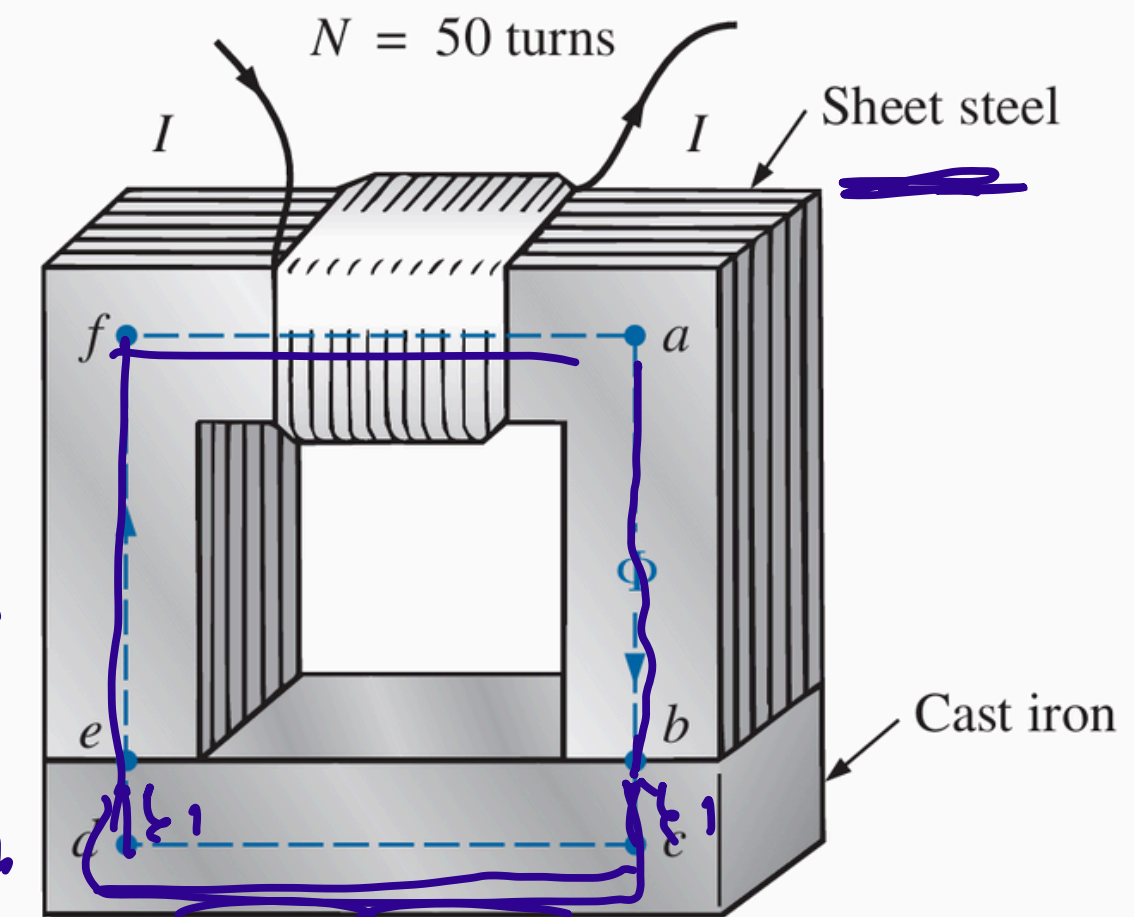
$$I = \frac{170 \times 0.16}{400}$$
$$= 68 \text{ mA}$$

EXAMPLE 12.1 For the series magnetic circuit in Fig. 12.12:

- Find the value of I required to develop a magnetic flux of $\Phi = 4 \times 10^{-4}$ Wb.
- Determine μ and μ_r for the material under these conditions.



EXAMPLE 12.2 The electromagnet in Fig. 12.14 has picked up a section of cast iron. Determine the current I required to establish the indicated flux in the core.



$$NI = H_{abfe} l_{abfe} + H_{edcb} l_{edcb}$$

$\downarrow$ $\downarrow$
 12 inch 5 inch

$$B_{ss} = \frac{\Phi}{A} = \frac{3.5 \times 10^{-4}}{6.452 \times 10^{-4}} = 0.542$$

$= 304.8 \times 10^{-3} \text{ m}$
 $127 \times 10^{-3} \text{ m}$

$$B_{ci} = 0.542$$

$$A = 1 \text{ inch}^2 = 6.452 \times 10^{-4} \text{ m}^2$$

$$\left. \begin{array}{l} H_{abfe} = 70 \\ H_{edcb} = 1600 \end{array} \right\}$$

$$I = \frac{70 \times 304.8 \times 10^{-3} + 1600 \times 127 \times 10^{-3}}{500}$$

$$l_{ab} = l_{cd} = l_{ef} = l_{fa} = 4 \text{ in.}$$

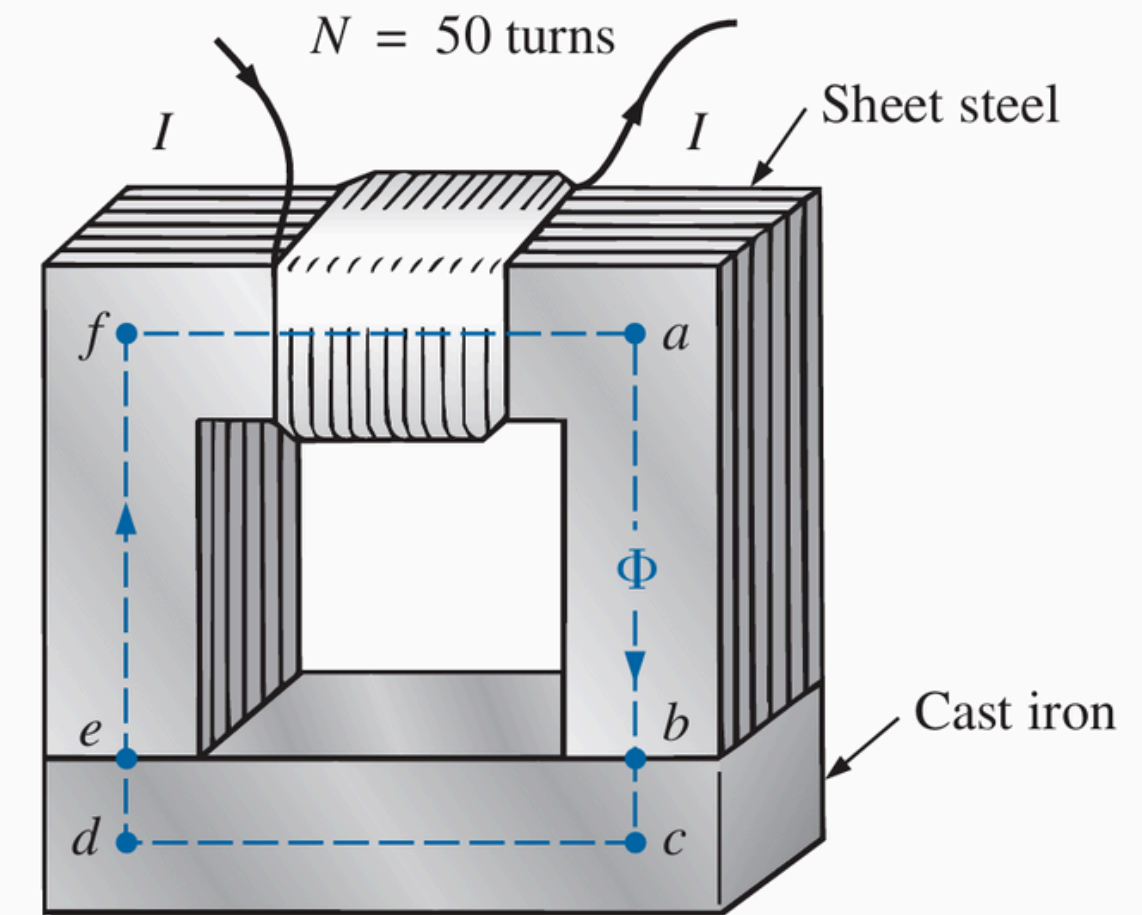
$$l_{bc} = l_{de} = 0.5 \text{ in.}$$

$$\text{Area (throughout)} = 1 \text{ in.}^2$$

$$\Phi = 3.5 \times 10^{-4} \text{ Wb}$$

EXAMPLE 12.2 The electromagnet in Fig. 12.14 has picked up a section of cast iron. Determine the current I required to establish the indicated flux in the core.

$$I = 4.49 \text{ A}$$



$$l_{ab} = l_{cd} = l_{ef} = l_{fa} = 4 \text{ in.}$$

$$l_{bc} = l_{de} = 0.5 \text{ in.}$$

$$\text{Area (throughout)} = 1 \text{ in.}^2$$

$$\Phi = 3.5 \times 10^{-4} \text{ Wb}$$

AGENDA



EXAMPLE 12.3 Determine the secondary current I_2 for the transformer in Fig. 12.16 if the resultant clockwise flux in the core is 1.5×10^{-5} Wb.

$$N_1 I_1 - N_2 I_2 = H_{abed} l_{abed}$$

$$B = \frac{\Phi}{A}$$

$$= \frac{1.5 \times 10^{-5}}{0.15 \times 10^{-3}}$$

$$= 0.1 \text{ T}$$

$$H = 20 \text{ A/m}$$

$$60 \times 2 - 30 \times I_2 = 20 \times 0.16 \Rightarrow 30 I_2 = 120 - 20 \times 0.16$$

$$I_2 = \underline{\underline{3.87 \text{ A}}}$$

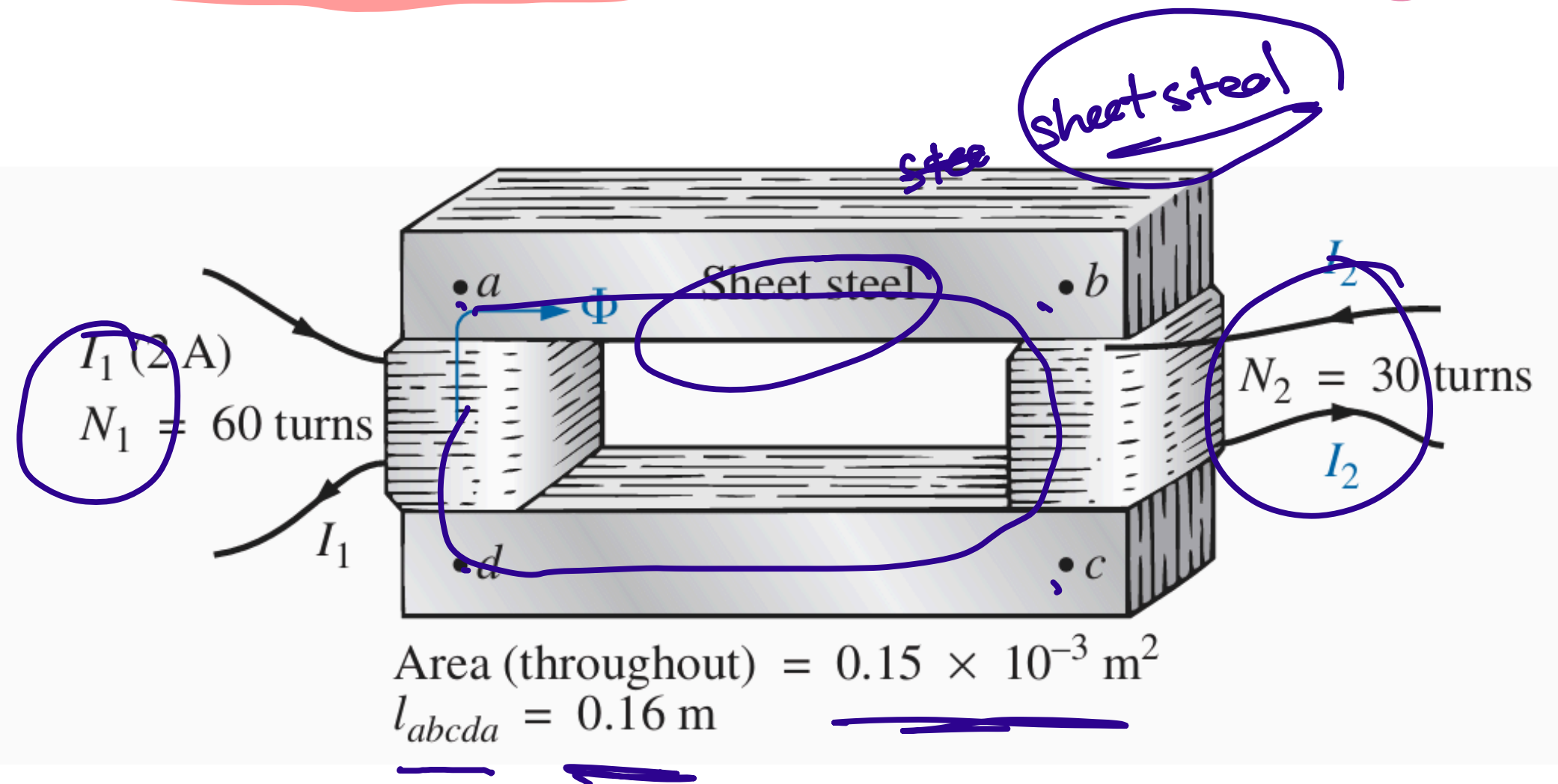


FIG. 12.16

Transformer for Example 12.3.

EXAMPLE 12.3 Determine the secondary current I_2 for the transformer in Fig. 12.16 if the resultant clockwise flux in the core is $1.5 \times 10^{-5} \text{ Wb}$.

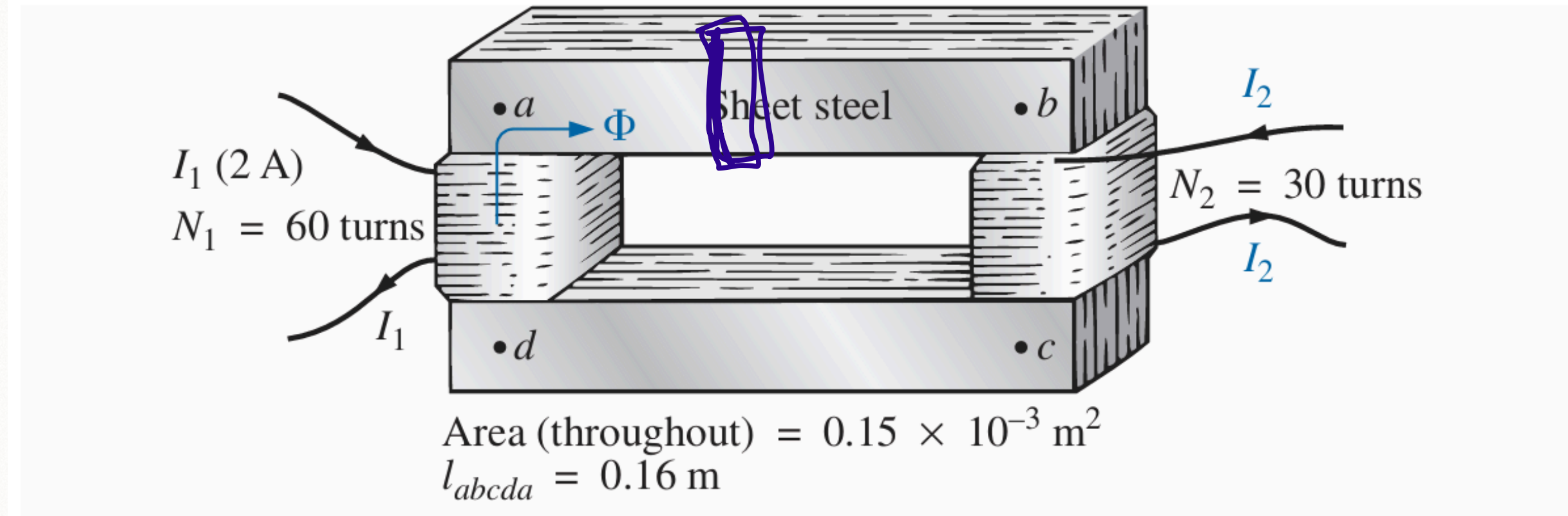


FIG. 12.16

Transformer for Example 12.3.

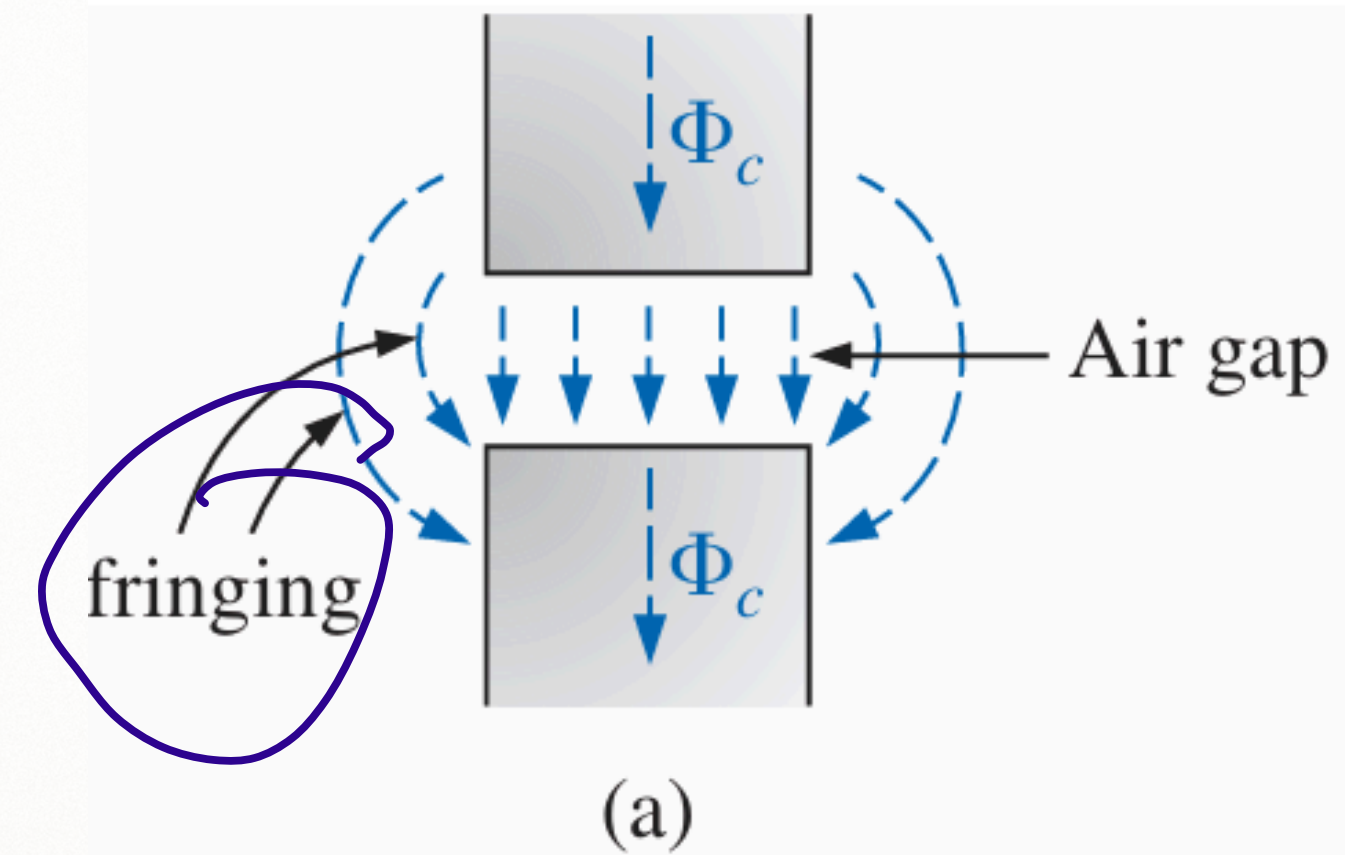
AGENDA



AGENDA



AIR GAPS



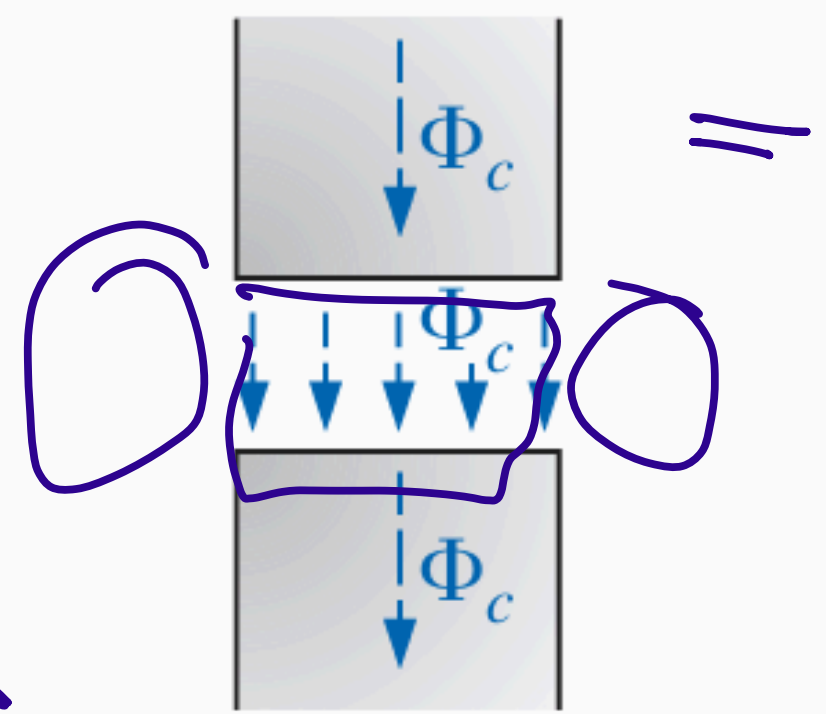
$$\boxed{\Phi_c} = \boxed{\Phi_e}$$

$$\Phi_c = \Phi_g$$

$$\Phi_c = \Phi_g$$

$$H$$

$$B = \mu H$$



$$B = \mu H$$

$$B = \mu_0 H$$

$$B = 4\pi \times 10^{-7} H$$

$$H_g = \frac{B}{4\pi \times 10^{-7}}$$

$$\mu_0 = (7.06 \times 10^{-5}) B$$

$$\frac{\Phi}{A} = B$$

EXAMPLE 12.4 Find the value of I required to establish a magnetic flux of $\Phi = 0.75 \times 10^{-4}$ Wb in the series magnetic circuit in Fig. 12.19.

$$NI = H_g l_g + H_l l_l$$

$$H_g = 7.06 \times 10^5 \times 0.5$$

$$= 3.98 \times 10^5 \text{ A}\cdot\text{t/m} \quad N = 200 \text{ turns}$$

$$H_l = 280$$

$$H_l l_l = 280 \times 100 \times 10^{-3}$$

$$= 28$$

$$H_g l_g = 3.98 \times 10^5 \times 2 \times 10^{-3}$$

$$= 796$$

$$NI = 796 + 28$$

$$I = 4.12$$

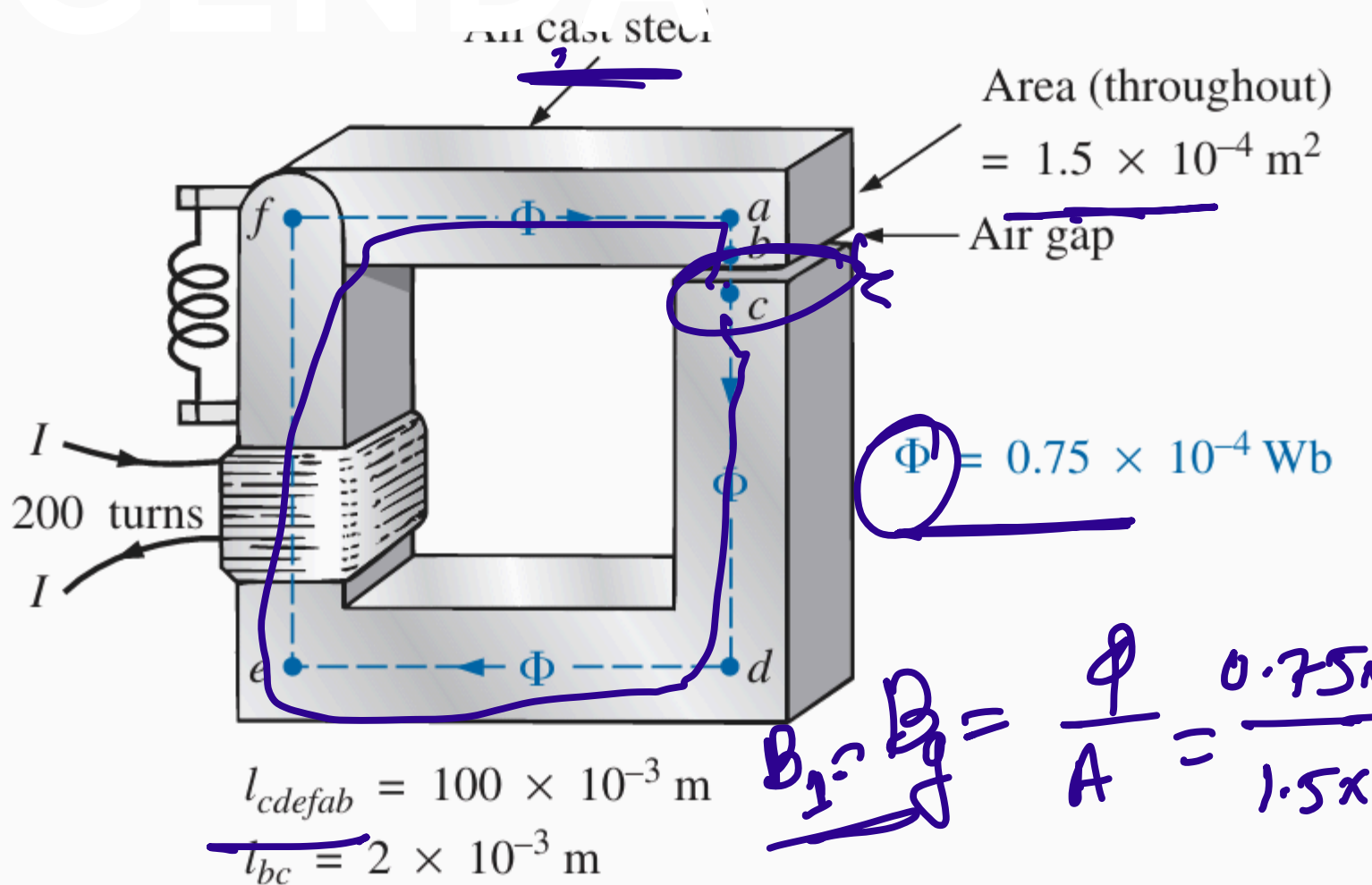


FIG. 12.19

Relay for Example 12.4.

$$N = 200$$

EXAMPLE 12.4 Find the value of I required to establish a magnetic flux of $\Phi = 0.75 \times 10^{-4} \text{ Wb}$ in the series magnetic circuit in Fig. 12.19.

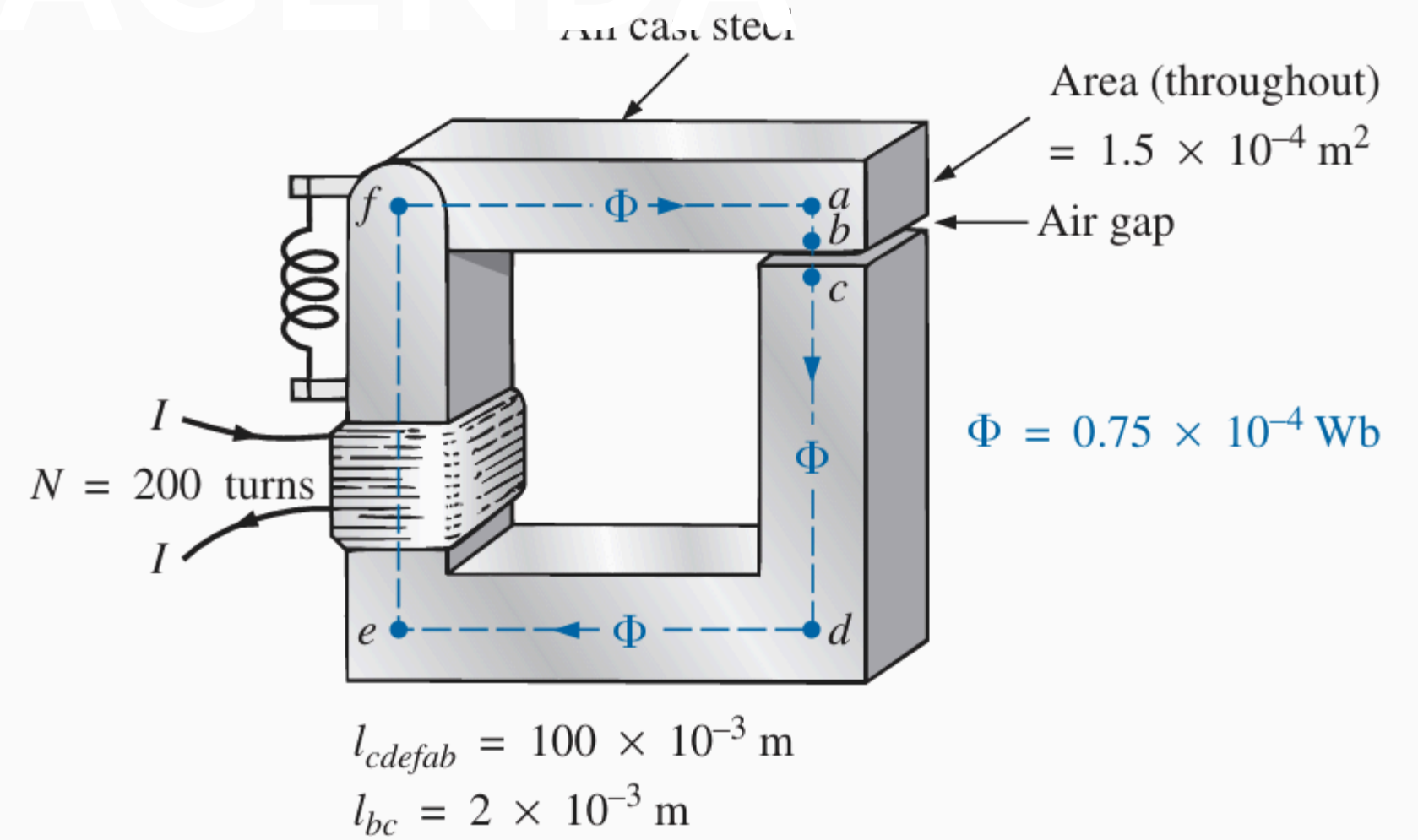


FIG. 12.19

Relay for Example 12.4.

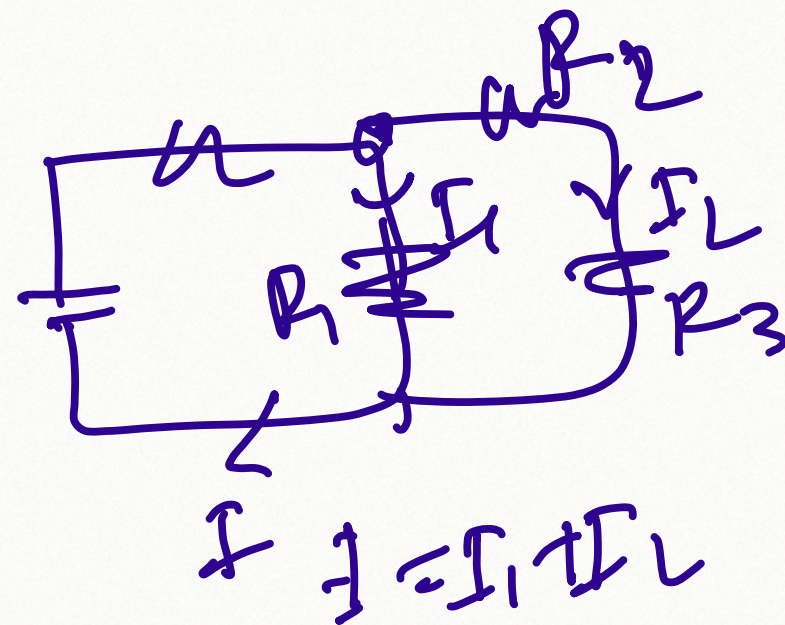
AGENDA



AGENDA



SERIES PARALLEL MAGNETIC CIRCUIT



$$I_1 R_1 = I_2 (R_2 + R_3)$$

$$I_2 (R_2 + R_3)$$

EXAMPLE 12.5 Determine the current I required to establish a flux of 1.5×10^{-4} Wb in the section of the core indicated in Fig. 12.21.

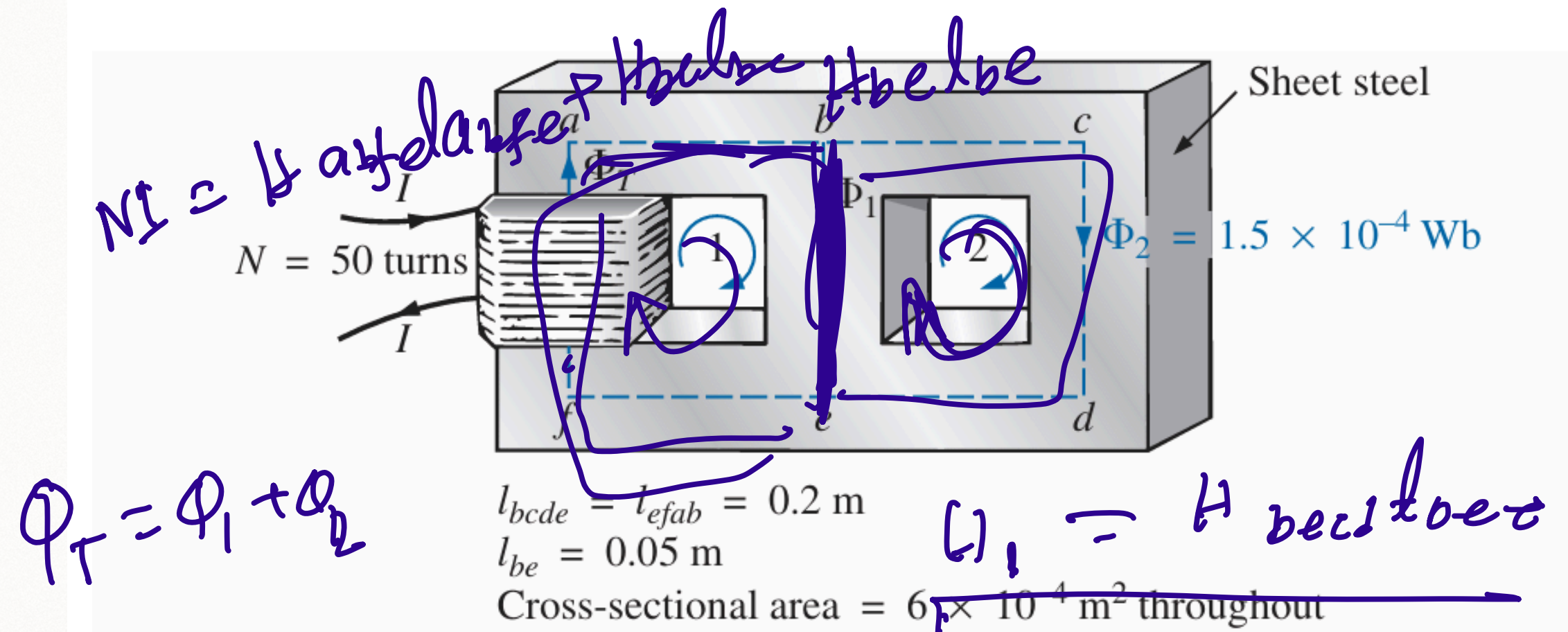
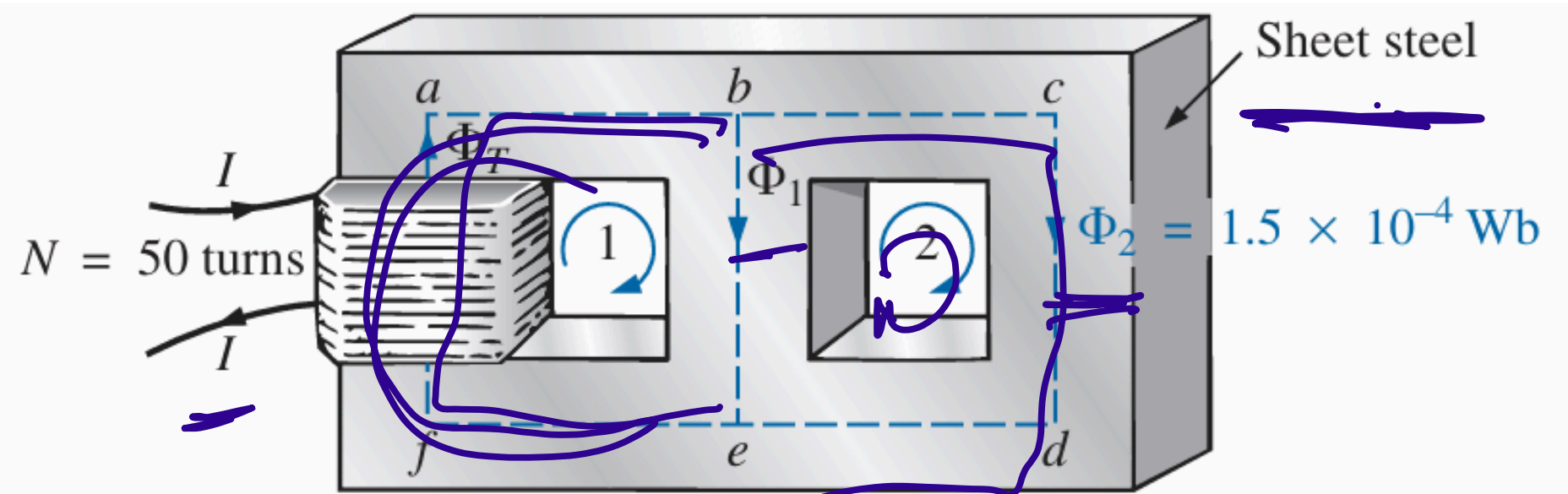


FIG. 12.21

Example 12.5.

EXAMPLE 12.5 Determine the current I required to establish a flux of 1.5×10^{-4} Wb in the section of the core indicated in Fig. 12.21.



$$l_{bcde} = l_{efab} = 0.2 \text{ m}$$

$$l_{be} = 0.05 \text{ m}$$

Cross-sectional area = $6 \times 10^{-4} \text{ m}^2$ throughout

FIG. 12.21

Example 12.5.

$$NI = H_a l_{bcde} + H_b l_{de}$$

$$NI = 400 \times 0.2 + 160 \times 0.05$$

$$\therefore I = 1.76$$

$$B = \frac{\Phi_2}{A} = \frac{1.5 \times 10^{-4}}{6 \times 10^{-4}} = 0.25 \text{ T}$$

$$H \Rightarrow 40 \text{ A/m}$$

$$B_2 = \frac{\Phi_2}{A} \therefore \Phi_2 = B_2 \times A$$

$$H_b l_{be} = 40 \times 0.2$$

$$H_{be} = \frac{40 \times 0.2}{0.05}$$

$$B_2 \quad B_{be} = 0.07 = 160 \text{ A/m}$$

$$\Phi_2 = 0.97 \times 6 \times 10^{-4}$$

$$= 5.82 \times 10^{-4}$$

$$\Phi_T = \Phi_1 + \Phi_2$$

$$= (5.82 + 1.5) \times 10^{-4}$$

$$= 7.32 \times 10^{-4}$$

$$B_T = \frac{\Phi_T}{A} = \frac{7.32 \times 10^{-4}}{6 \times 10^{-4}} = 1.22$$

$$H_{ref} = 400$$

AGENDA



DETERMINING Φ

EXAMPLE 12.7 Find the magnetic flux Φ for the series magnetic circuit in Fig. 12.24 for the specified impressed mmf.

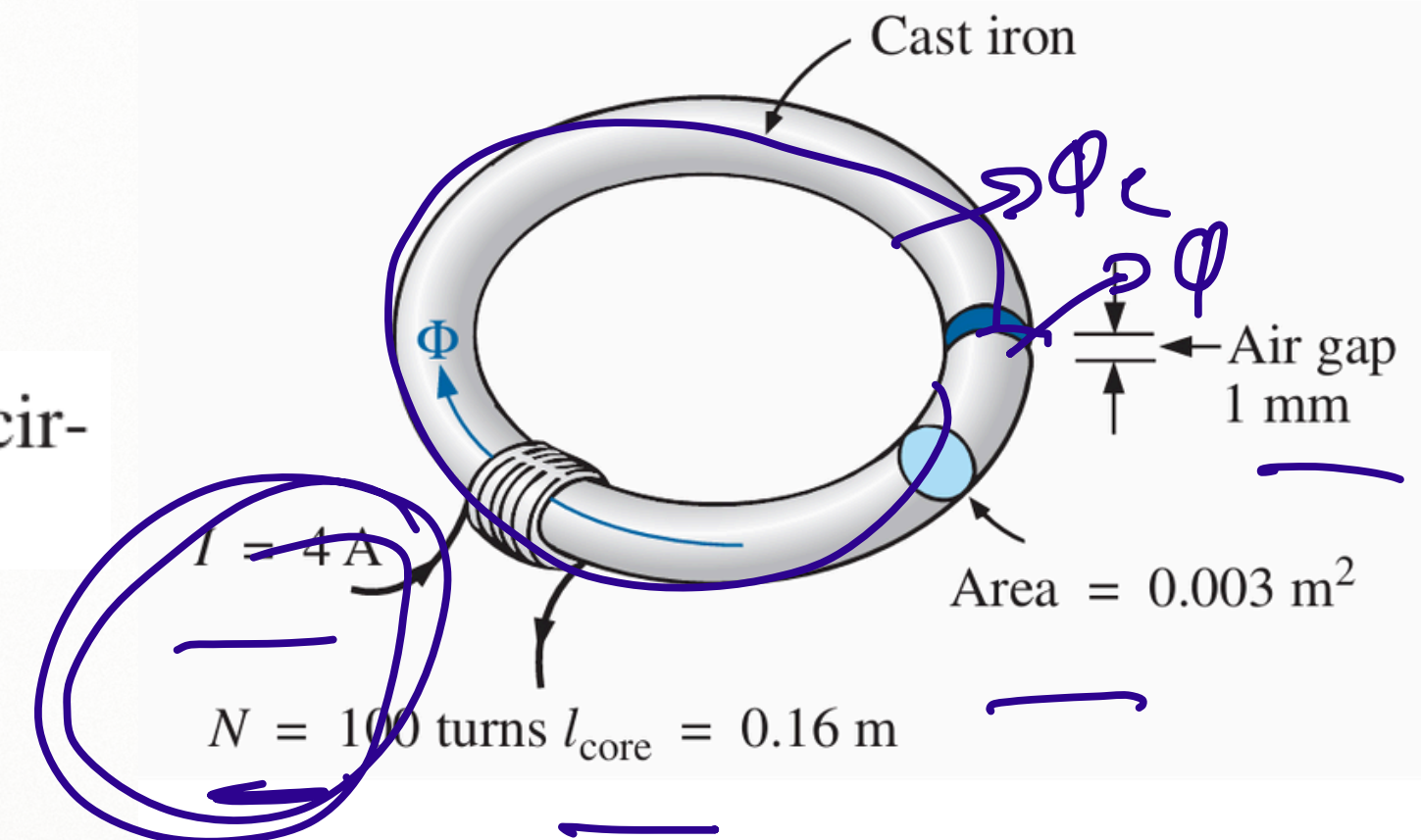


FIG. 12.24
Example 12.7.

$$B = \frac{\Phi}{A}$$

$$\begin{aligned} & \text{Handwritten: } \textcircled{NI} = 400 \\ & 4 \times 100 = H_g \times 10^{-3} \end{aligned}$$

$$H_g = \frac{400}{10^{-3}} = 4 \times 10^5 \text{ At/m}$$

$$\begin{aligned} B_g &= \mu H_g = 4\pi \times 10^{-7} \times 4 \times 10^5 \\ &= 0.503 \text{ T} \end{aligned}$$

$$\Phi_g = \Phi_c = B_g \times A = 0.503 \times 0.003 = 1.51 \times 10^{-3} \text{ Wb}$$

$$\begin{aligned} B_g &= \mu H_g \\ &= 4\pi \times 10^{-7} \end{aligned}$$

$$\begin{aligned} & \text{Handwritten: } \textcircled{NI} = 400 \\ & \text{Handwritten: } \cancel{H_c l_c} + H_g l_g \\ & \text{Handwritten: } 400 \end{aligned}$$

$$H_c = 1500 \text{ (B-H curve)}$$

$$H_{lc} + H_{lg} \rightarrow 640$$

$$= 1500 \times 0.16 + 0.34 \times 10^5 \times 10^{-3}$$

$$= 640$$

$$\frac{640 - 4000}{640} = 37.5\%$$

$$\Phi = 0.7 \times 1.57 \times 10^{-3} \\ = 1.057 \times 10^{-3}$$

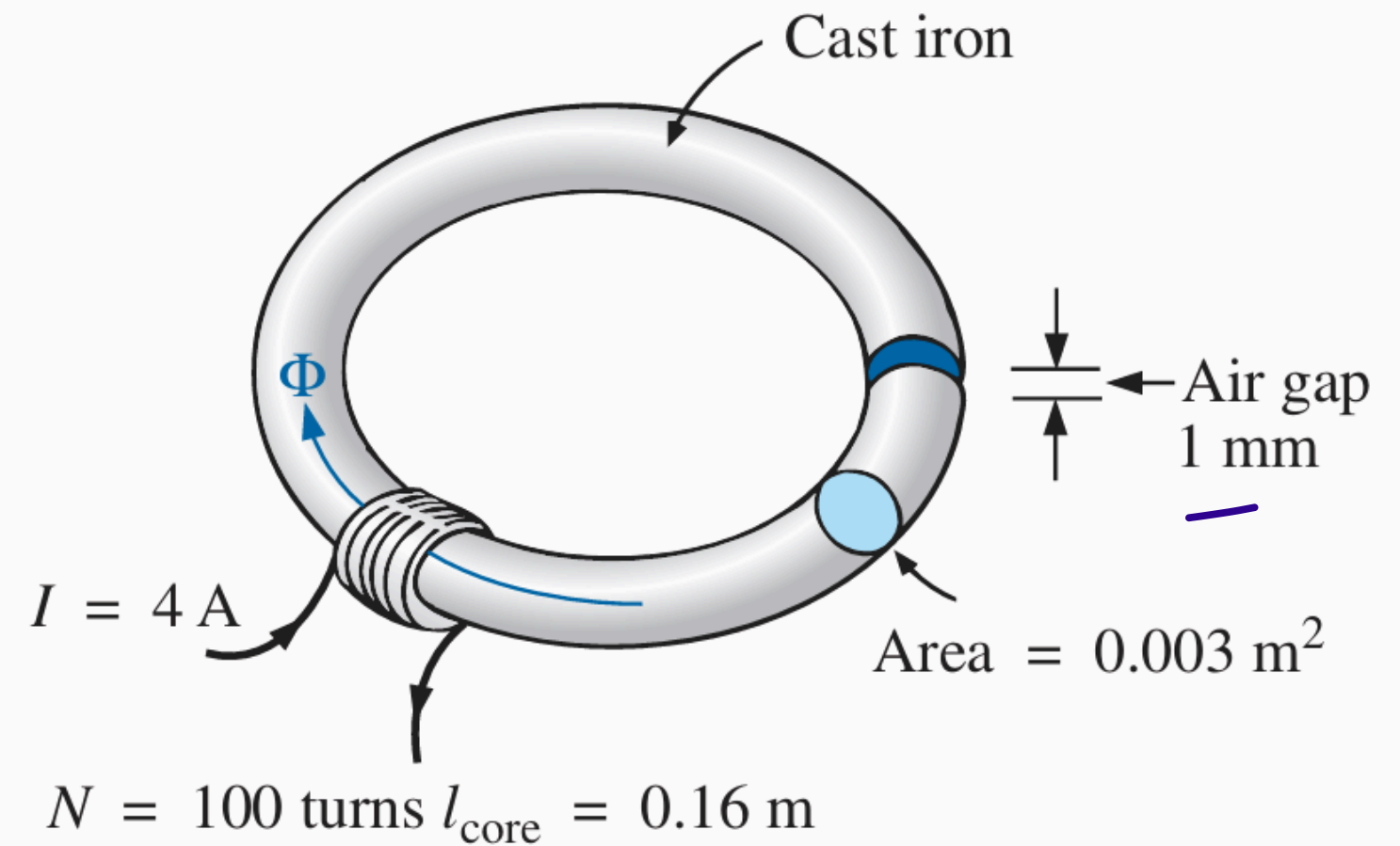


FIG. 12.24

Example 12.7.

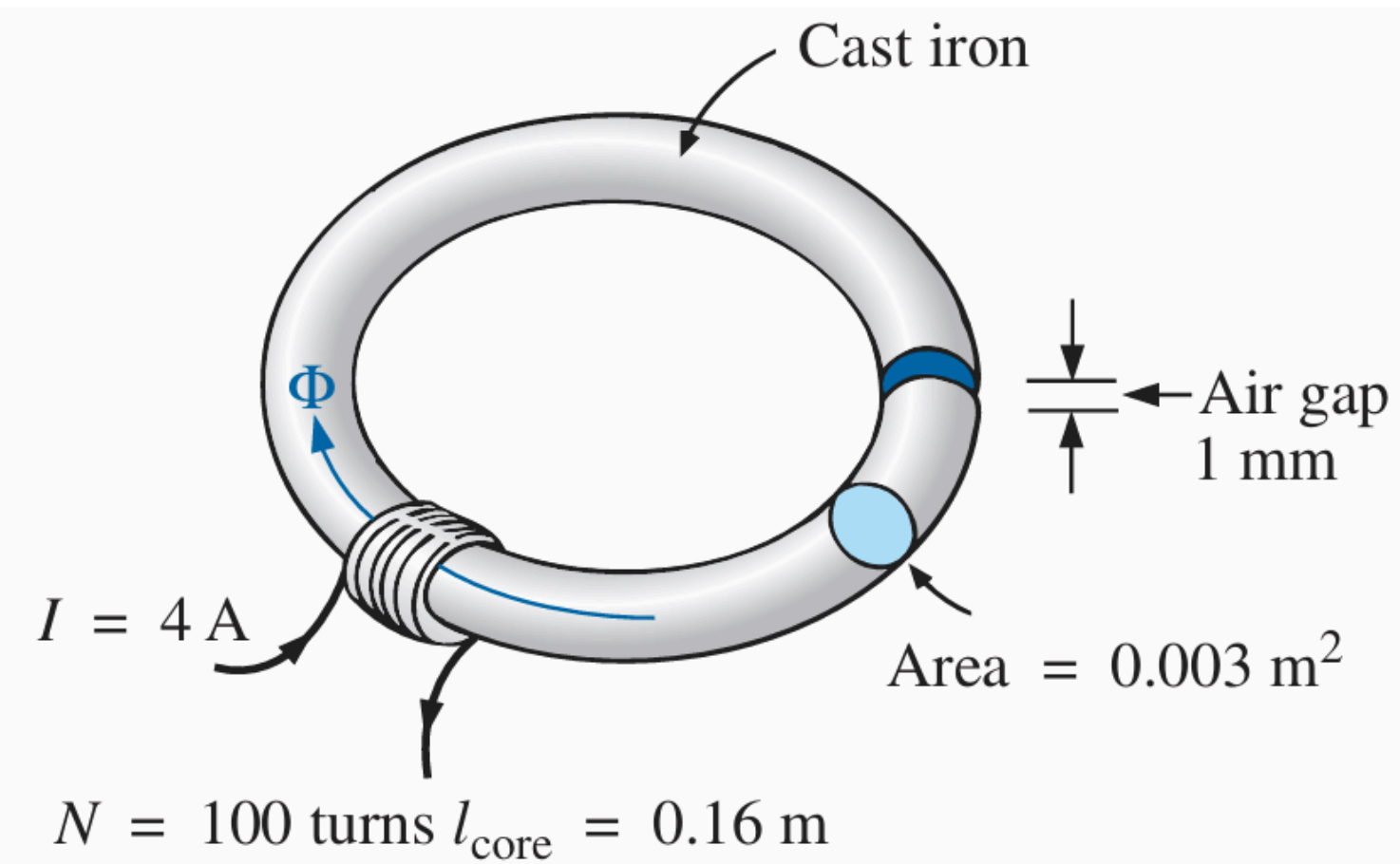


FIG. 12.24

Example 12.7.

$$H_{\text{Fe}} + H_{\text{g}} = 136 + 280 = 416$$

416

57

$$B = \frac{\Phi}{A} = \frac{1.057 \times 10^{-3}}{0.003} = 0.352 \text{ T}$$

$$H_{\text{Fe}} = 850$$

$$H_{\text{Fe}} l_{\text{Fe}} = 850 \times 0.16 = 136$$

$$H_{\text{g}} l_{\text{g}}$$

$$H_{\text{g}} = \mu B_{\text{g}}$$

$$= 4\pi \times 10^{-7} \times 0.352$$

$$= H_{\text{g}} l_{\text{g}} = \boxed{} \times 10^{-3} = 280$$

19. Find the magnetic flux Φ established in the series magnetic circuit in Fig. 12.46.

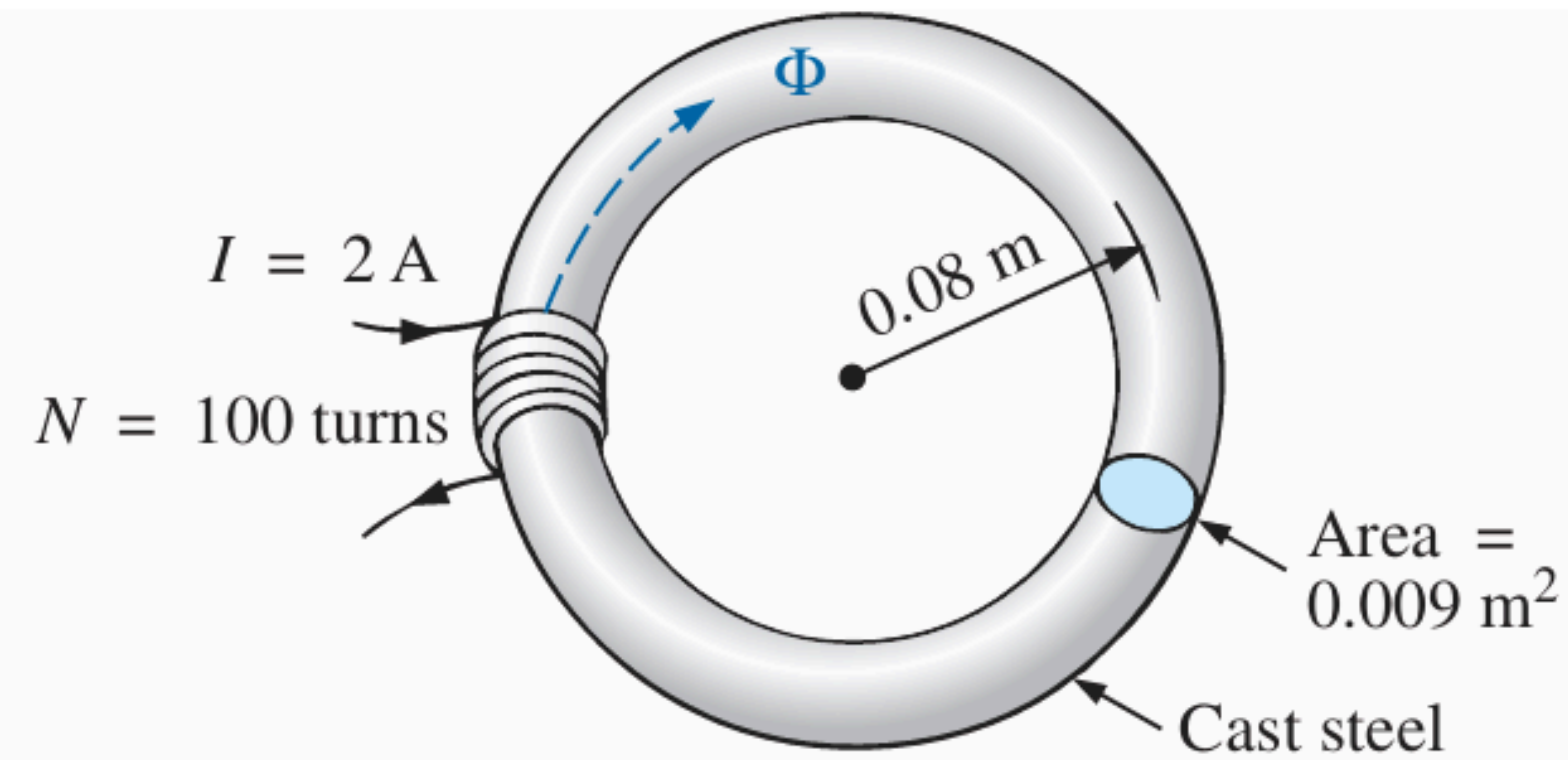


FIG. 12.46

Problem 19.